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# Dielectric characterization analysis of natural fiber based hybrid composite for microwave absorption in X-band frequency

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**Abstract**
Emerging issues on electromagnetic interference (EMI) caused by blooming of new electronic gadgets and its implementation in different sectors. Thus, the minimization or to control the effect EMI on electronic devices and the designing of stealth equipment requires special attention for dielectric materials of different blends. To modify the dielectric properties of any material, it requires combination of more than one compatible constituent. The designing of hybrid composite with optimum wt.% contributes good dielectric substrate for interaction with the microwave. The thin layer with 3 mm thickness hybrid composite of banana stem and rice husk dust (BAN/RH) with particle size 150 µm in ratio 50–50 wt.% has been synthesized for X-band testing specimen. The results of microwave properties with reflection loss − 14.01 dB attenuates 96.02% of incident power of EM wave significantly describes the importance of dielectric properties of single layer hybrid composite material in X-band frequency range.

**Keywords** Hybrid composite · Electromagnetic wave · Dielectric · Microwave absorber · Shielding properties

## 1 Introduction

With the rapid evolution of communication technologies and the widespread utilization of electronic devices, the inadvertent proliferation of electromagnetic (EM) signals is anticipated to become a significant form of pollution by 2030 [1]. The EM pollution caused by interference of unwanted EM signal can be controlled by using a special kind of material or thin coating which has variable dielectric property with different frequency range [2]. Currently, cutting-edge research is focused on the synthesis of microwave absorbers which is a vital component in stealth technology, communication sectors, and information data processing centers. The functionalization of this microwave absorber at micro scale requires desirable dielectric properties if the raw material is non-metallic and both dielectric and magnetic properties if the raw material is metallic. The dielectric characteristics play an important role in evaluating the microwave response performance of absorbers, as they are dependent on the physicochemical properties of the fabrication of material [3]. The real and imaginary components of complex permittivity describe the performance of the microwave absorber storing of EM energy as strength of material and the ability of converting EM energy in to the dissipated thermal energy [4]. Similarly, the tangent loss confirms the attenuation ability of the material in presence of EM wave propagate through it. Thus, dielectric constant, dielectric loss and other related parameter describing the performance of microwave absorbers are of significant contribution in analysis of microwave interaction with the material. The recent commercial microwave absorber prepared from mixture of polyurethane and polystyrene are not eco-friendly and also high cost for fabrication [5]. Further, these materials are basically composed of carbon coating which is not 100% ecofriendly from an environmental point of consideration. Thus, the selection of appropriate alternative potential material for fabrication of microwave absorber creates special attention for carbonaceous composition which is abundant in plant-based biomaterial. The chemical compositions of biomaterial are basically associated with different lingo cellulosic material such as cellulose; hemicellulose and lignin which are basically responsible for not only maintaining the stability of the composite material but

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also create the huge amount of activated carbon [6]. Further, the resistive nature of carbon which easily transforms the current to the thermal energy facilitates the absorption phenomena in microwave absorbing materials (MAM) with release of thermal radiation to the environment. As most of the work related to synthesis of MAM consists of a single organic material which may not possesses the properties like exceptional mechanical strength, light weight, chemical and thermal stability, required for aerospace application in stealth industries, so there is need of heterogeneous material which may improves in the desired characteristics. The present work encompasses a novel approach to minimize electromagnetic interference (EMI) through the development of a hybrid composite, blending banana stem and rice husk dust in a 50–50 wt.% ratio. The amount of mixture is considered after doing stoichiometric calculations for both the epoxy polymer matrix and filler material to ensure there is no error in chemical measurements. The fibrous nature of banana stem contributes to the overall structural integrity of the hybrid composite. Its cellulose-rich composition may enhance dielectric properties, influencing the material's ability to interact with electromagnetic waves [7–10]. Additionally, the removal of impurities and the increase in porosity during the treatment processes can positively impact the absorption and attenuation of electromagnetic energy. Rice husk dust, being a byproduct of rice processing, may introduce unique elements such as silica and lignin to the hybrid composite. These elements could potentially influence the material's conductivity and absorption characteristics. The porous nature of rice husk dust particles may contribute to creating an intricate structure within the composite, affecting the propagation and dissipation of electromagnetic waves [11–13]. The use of ultrasonic alkalization to modify the raw fibers enhances compatibility, removes impurities, and results in a structurally distinct composite with increased porosity [9]. Characterization techniques such as SEM, EDS, and FTIR provide a comprehensive analysis of morphological and elemental changes, offering insights into the material's electromagnetic properties. The study explores the frequency-dependent behavior of dielectric properties, conductivity, and reflection loss across the X-band frequency range. The impedance matching and the absorptivity, with a 2.13 GHz bandwidth, highlight the material's suitability for practical applications in radar absorbing material. This research contributes significantly to the field by offering a sustainable solution to EMI challenges, showcasing the potential of the hybrid composite in real-world applications.

### 1.1 Dielectric theory for computation of microwave parameters

The propagation of electromagnetic (EM) wave in any medium depends on the characteristic properties through which it propagates. The characteristics of a material's dielectric and magnetic properties have a significant impact on the space and time-varying aspects of a propagating wave. The interaction of EM wave with material may be approached either by macroscopic behavior by analysis of Maxwell's equations or by microscopic performance of waves with material structure [14]. For an excellent and potential microwave absorber, the material should possess well matched impedance and strong attenuation capacity which is strongly influenced by dielectric properties. As the electrical characteristics of the composite materials are mainly influenced by the movement of charge particle due to dielectric polarization with a good dielectric property, the velocity of EM wave also controlled by dielectric constant [15]. The materials having low dielectric constant permits the EM wave easily and vice-versa. The creation and annihilation of bound charge and free charge due to interaction of incident electric field E on the material realigned the charged species of polar groups in every domain of the material which enhances the energy storage. This energy storing ability differentiates the behavior of the material from that of vacuum by the specific property called relative dielectric constant $\varepsilon'$.

Considering Maxwell's EM wave equation in synthesized dielectric material is given as:

$$\nabla^2\vec{E} - \mu\varepsilon\frac{\partial^2\vec{E}}{\partial t^2} = 0 \tag{1}$$

where $\mu$ is magnetic permeability and $\varepsilon$ is the dielectric permittivity. Since the composite have very low magnetic properties, the real part of magnetic permeability is assumed as unity ($\mu' = 1$) and the imaginary magnetic permeability is assumed as zero ($\mu'' = 0$). Equation (1) reduces to

$$\nabla^2\vec{E} - \varepsilon\frac{\partial^2\vec{E}}{\partial t^2} = 0 \tag{2}$$

Microwave absorption analysis of any dielectric material synthesized from organic material are assessed by three factors as the permittivity ($\varepsilon$), the material reflection and the material absorption factors.

$$\varepsilon = \varepsilon' - j\varepsilon'' \tag{3}$$

where, $\varepsilon$ is the dielectric permittivity which arises by induced polarization in the material, $\varepsilon'$ is the real part which is responsible for storing EM wave inside the material medium, $\varepsilon''$ is the imaginary part which indicates the loss of energy in EM wave by dissipating into heat energy and $j$ is a complex number ($j = \sqrt{-1}$) [16]. Dielectric loss tangent measurement is another metric for determining the inherent quality of composite as an absorbing material that results in EMW absorption and is given as

$$\tan\delta = \frac{\varepsilon''}{\varepsilon'} \tag{4}$$

where, tan$\delta$ is tangent loss and $\delta$ is angle of loss. Wave attenuation increases with increasing material loss tangent. For materials with low loss, $\delta \ll 1$ and tan $\delta \approx \delta$. Thus, the electrical conductivity of dielectric material which is given by:

$$\sigma = \omega\varepsilon_0\varepsilon'' = 2\pi f\varepsilon_0\varepsilon'\tan\delta \tag{5}$$

where, $\omega$ is angular frequency,$\varepsilon_0$ is the permittivity of the free space, $f$ is frequency. For a lossy dielectric material as $\sigma \neq 0$, then propagation constant of the material is given by:

$$\gamma = \sqrt{j\omega\mu(\sigma + j\omega\varepsilon)} = \alpha + j\beta \tag{6}$$

where the real part $\alpha$ is associated with attenuation of wave and the imaginary part $\beta$ is associated with propagation of wave.

$$\beta = \frac{\omega\mu\sigma}{2\alpha} \Rightarrow \alpha = \omega\sqrt{\frac{\mu\varepsilon}{2}}\sqrt{1 + \frac{\sigma^2}{\omega^2\varepsilon^2}} - 1 \tag{7}$$

The transmission coefficient T in a lossy dielectric determines the transmitted electric field $E_t$ in a material.

$$E_t = TE_0e^{j(\omega t - \gamma Z)} \tag{8}$$

Solving Eqs. (6), (7) and (8) we get

$$(1 + \phi) = T$$

$$\phi = \frac{Z_{in} - Z_0}{Z_{in} + Z_0} \tag{9}$$

Since reflection loss (RL) is the magnitude of reflection coefficient $\phi$ in dB, so RL is given by RL = $20\log_{10}|\phi|$

$$= 20\log_{10}\left|\frac{Z_{in} - Z_0}{Z_{in} + Z_0}\right| \tag{10}$$

where, $Z_0$ is the EM wave impedance of the free space i.e. 377 Ω and $Z_{in}$ is the input EM wave impedance is given by

$$Z_{in} = Z_0\sqrt{\frac{\mu_r}{\varepsilon_r}}\tanh\left(j\frac{2\pi fd}{c}\sqrt{\mu\varepsilon}\right) \tag{11}$$

where, $f$ and $d$ are frequency and thickness of material [17].

## 2 Materials and methods

### 2.1 Materials

The sample for hybrid material to be fabricated is basically composed of 50–50 wt.% of banana stem and rice husk dust. Raw materials were dried under sunlight for 4–6 days before being treated with a 1N NaOH solution. Following the process of ultrasonication the dried raw fibers are submerged in NaOH solution in a beaker and the beaker was placed inside a bath sonicator for proper blending of NaOH in the subatomic parts of fiber cellulose. The proper reason of considering NaOH is due to its intrinsic properties which increase fiber swelling, expanding their surface area and increasing porosity for improved penetration of surfactant. This process, in turn, increases the tensile strength and wear resistance of fibers by enhancing cellulose crystallinity [18]. Mercerization with NaOH also imparts luster to fibers, changing their light-reflecting properties, and boosts absorbing capacity. Additionally, the treatment ensures improved dimensional stability, converting treated fibers less susceptible to shrinkage or elongation in response to variations in moisture or temperature [11–13, 18]. The sonicator was set to 125 kHz frequency and the process was continued for 1 h. The pH of the treated fibers is maintained by cleaning them in distilled water thrice. The treated fibers again placed for 2 h in hot air oven at temperature of 60 °C for removal of moisture if any to facilitate the fiber materials for better grinding into particles of 150 µm size using a test sieve. The general hand layup method was used to fabricate the composite by preparing blends of epoxy polymer and hardener in a ratio of 10:1, which were mixed for 30 min using a mechanical stirrer. The powder of banana stem and rice husk were then added to the polymer matrix in separate beakers and stirred for 15 min for proper mixing. For pre preparation of mold, the mold was covered with a silicon sheet to maintain a plain surface texture. Silicon oil was sprayed on the Teflon sheet to avoid sticking of the resin. The fiber/resin mixture was poured into a 15 cm × 15 cm × 1 cm mold. The mold was coved and C-clamps were used to tighten. After 12 h of curation the composite was demolded and with the help of a jigsaw cutter required dimensions (23 mm × 11 mm × 3 mm) of composite was cut for microwave test specimen (Fig. 1).

The microwave measurement of the BAN/RH hybrid composite has been conducted over a wide band of frequency for measurement of complex dielectric permittivity, material reflection and absorption factors. The dielectric constant and loss factor were measured with two port vector network analyzers. A pair of horn antennae for transmission and reception of EM signal has been performed by free space measurement method in which material under testing was mounted over a table. With the proper calibration of the set up with air/water within the X-band frequency the instrument kept ready for testing of sample.

### 2.2 Characterization of BAN/RH hybrid composite

Both the raw and synthesized material are characterized using techniques such as SEM, EDS, and FTIR to investigate the changes in treated fibers and their composites and their

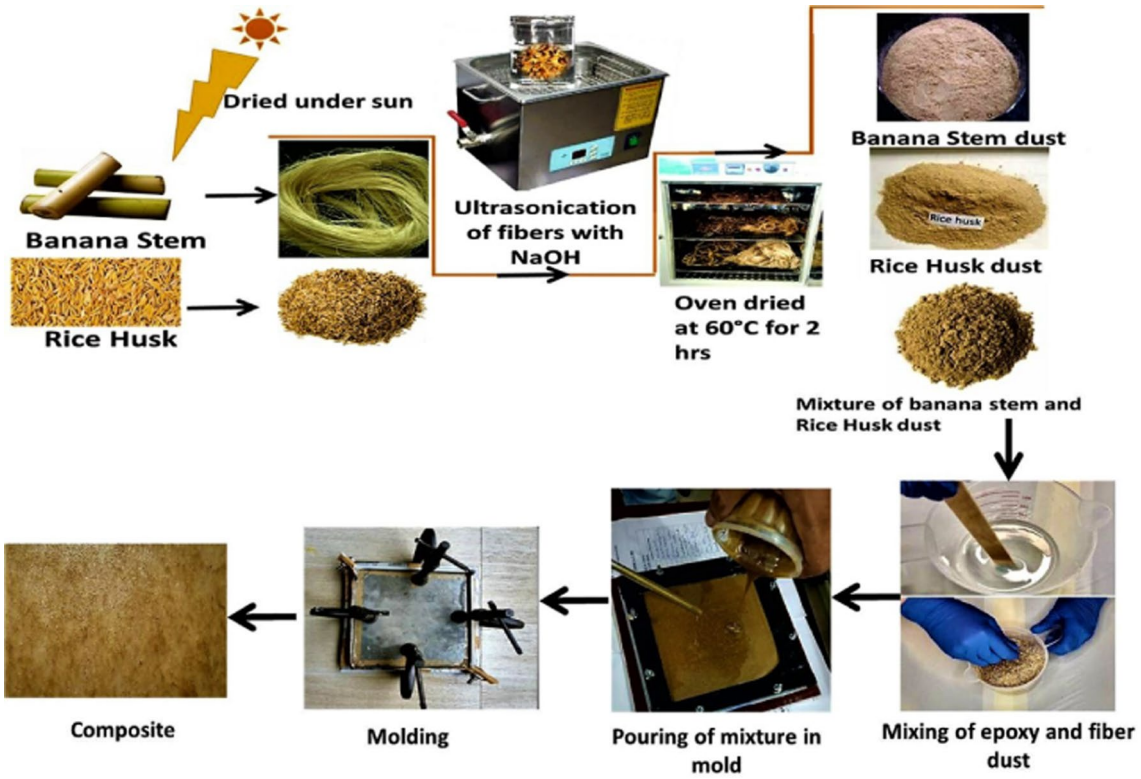



**Fig. 1** Preparation of BAN/RH hybrid composite

correlation with electromagnetic wave (EMW) absorption capacity. Figure 2 shows the morphological changes that occurred on the SEM images of BAN/RH hybrid composites operating with a voltage of 5 kV. The surface of the sample was coated with a conductive gold tape after being scratched, and a current of 18 mA and a pressure of 0.1 torr were applied to make it conductive. As shown in Fig. 2a, the surface of the untreated BAN/RH hybrid composite is covered with impurities and foreign particles associated with groove-like structures on the surface of the fiber. The surfaces of the treated BAN/RH are free from impurities and foreign materials are shown in Fig. 2b. Impurities such as wax, lignin, cellulose, and hemicellulose are cleaned from the surface of the fiber by the action of surfactants, resulting in a distinguishable view of fibers with a higher number of porous structures, as shown in Fig. 2c [19]. The significant

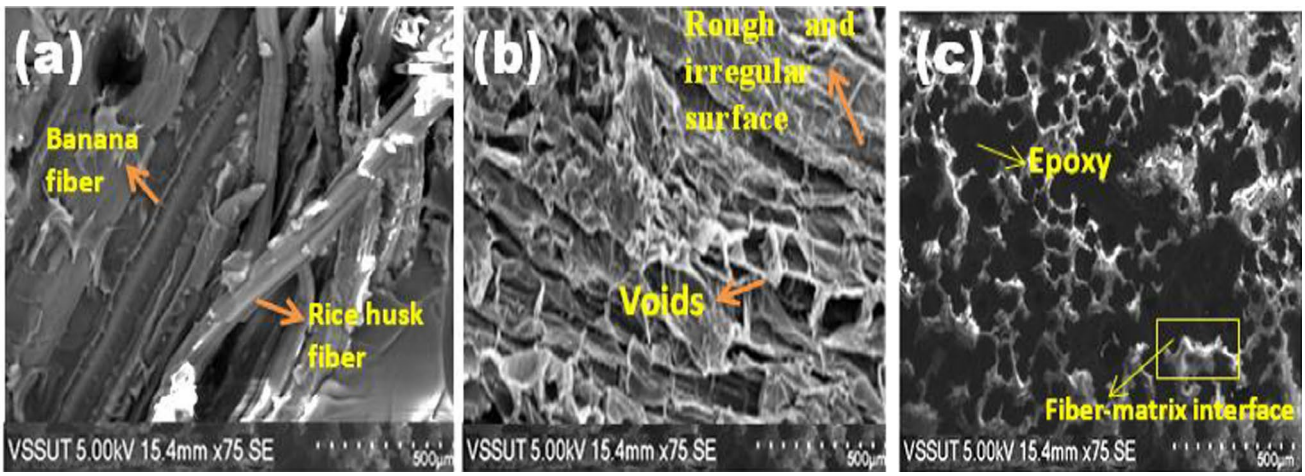



**Fig. 2** SEM of **a** untreated BAN/RH powder. **b** Treated BAN/RH powder. **c** BAN/RH hybrid composite

number of voids seen on the composite surface indicates that it has a high level of porosity. This porosity is a result of the surfactant's action due to removal of impurities from the fiber's surface. Moreover, due to removal of moisture the surface of the treated fiber is significantly hydrophobic in nature this creates large numbers of pores in the surface of the BAN/RH composite [7–9, 20–22].

The composition of the BAN/RH hybrid composite was analyzed through the use of energy dispersive X-ray spectroscopy (EDS). Figure 3 shows the EDS of a BAN/RH hybrid composite. The EDS data indicate the primary elements of treated composite are carbon and oxygen. The carbon fraction in composite was 67.80% that outperforms other natural fibers. Possible causes include the presence of non-cellulose materials in banana and rice husk [23]. Because hemicellulose, lignin, and other cellulosic constituents are removed from the fiber surface during alcohol treatment; the oxygen wt.% is less in comparison to the carbon wt.%. In the composite, there are traces of silicon other than oxygen and carbon. The existence of silicon improves the rigidity of the composite by improving fiber and matrix interaction, therefore raising the composite's tensile strength [20]. In the treated composite, a trace of 32.17% oxygen was detected in EDS of the treated BAN/RH hybrid composite. Suggesting the presence of functional group connected to oxygen, such as glyceraldehyde, which produces active absorption sites [21].

The spectrum of functional elements in Fig. 4 represents the comparison between an untreated and an alkali (NaOH) treated BAN/RH fiber. The primary constituents of lignocellulosic fibers, including cellulose, lignin, and hemicelluloses, contribute to the spectral bands observed. Therefore, these components are mainly responsible for the differences observed between the samples. The untreated fiber displays a band at 3307.58 $cm^{-1}$, which originates from the O–H stretching vibration in hemicellulose, cellulose, and absorbed moisture in the intracellular regions [22]. The presence of a peak at 1616 $cm^{-1}$ in the FTIR spectrum of the BAN/RH hybrid composite is indicative of the C–O stretching of the carbonyl groups in hemicellulose [7]. Following the application of alkali treatment to the BAN/RH fibers, the characteristic peaks of hemicellulose and lignin between 1723 to 1241 $cm^{-1}$ were absent in the FTIR spectrum. Upon mercerization, the intensity of the O–H absorption band at 3159 $cm^{-1}$ decreases suggesting a decrease in the hydroxyl O–H group within the fibers due to weak O–H stretching hydrogen bonds. This phenomenon is attributed to the disruption of the hydrogen bonds between cellulose and hemicellulose molecules [8]. The strong peak corresponding to the ester group in hemicellulose at 1622 $cm^{-1}$ disappeared, indicating the removal of hemicellulose and bonding materials from the fibers surface due to the surface treatment. The findings suggest that hemicellulose ester group is more susceptible to removal through alkali treatment. This treatment is also effective in removing a significant portion of uranic acid from the hemicellulose xylan, leading to absence of cellulose (C–O) stretching [9]. The presence of lignin in BAN/RH fibers is evidenced by absorbance bands at 1425 and 1371 $cm^{-1}$, corresponding to its asymmetric and symmetric deformations, respectively. Although NaOH treatment preferentially targets hemicelluloses in fibers, it effectively removes lignin and hemicelluloses components in hybrid fibers. Further, elimination of the organic crusts increases the hydrophobic behaviour of the fibers.

## 3 Results and discussion

When an electromagnetic wave encounters a medium composed of polymeric material reinforced with bio fibers, the random orientation of various atomic and molecular carriers will migrate, creating heterogeneous interfaces that generate

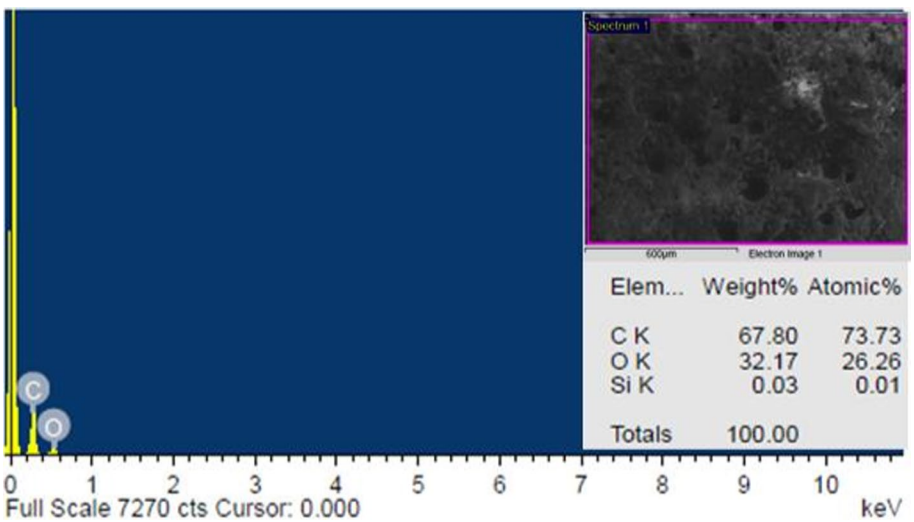



**Fig. 3** EDS of BAN/RH hybrid composite

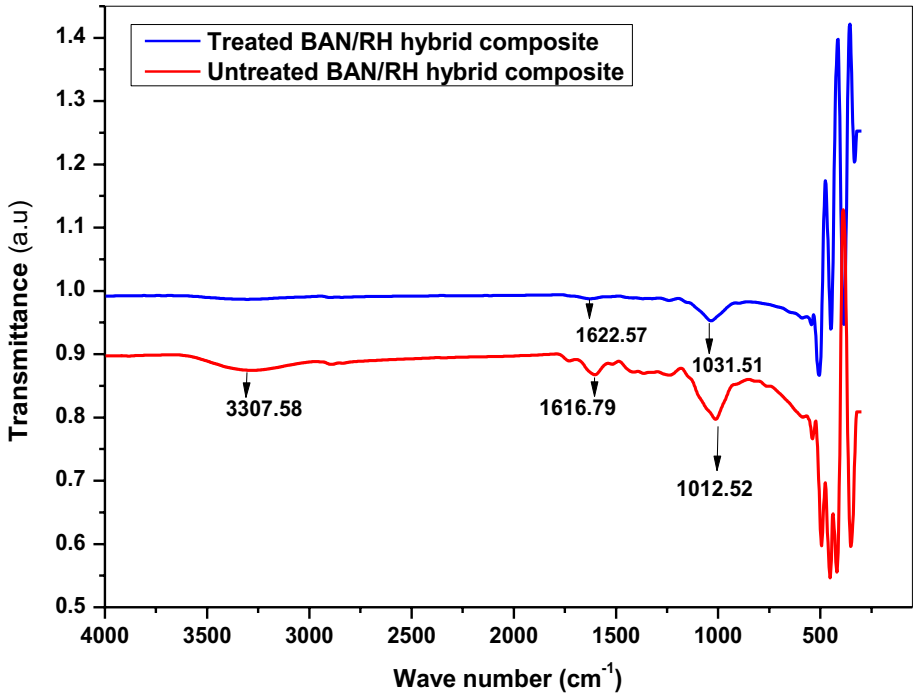



**Fig. 4** FTIR spectra of BAN/RH hybrid composite

interfacial polarizations. The absorption of energy in this material is associated with different processes, including dielectric loss due to the polarization of dipoles created by ionic conduction and the relaxation mechanism of charge carriers [7, 8]. Loss also occurs due to conduction caused by the flow of free electrons aligning themselves with the incident wave. In this case, the dielectric material is a polymer-blended bio composite, and its atomic structure allows for the reorientation of dipoles, creating a net average dipole moment per molecule [10]. The electromagnetic field can generate heat in the material through interaction with its structure on both the molecular and electronic levels. This interaction converts electromagnetic wave energy into heat energy within the material, resulting in heat absorption [7]. Multiple reflections are acknowledged by changing the materials structures will cause the EM waves to expand and additionally further develop the absorption capacity of the EMW absorbers as schematically shown in Fig. 5a, b.

The intrinsic characteristics of the composite material like complex permittivity is measured by two port vector network analyzers (VNA). Measurements were taken for both the real and imaginary components of permittivity at X-band frequency (8.2–12.4 GHz). The composite specimen was subjected to free space measurements using horn

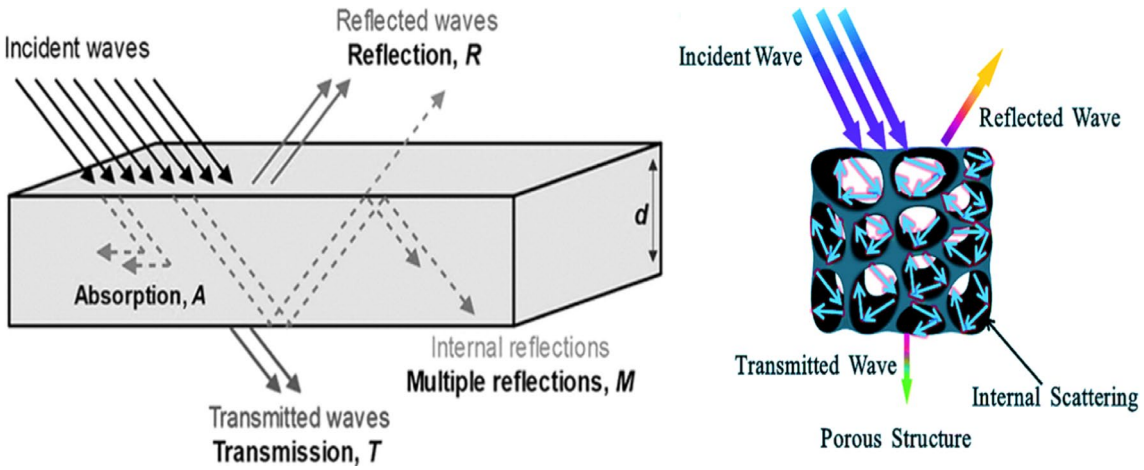



**Fig. 5** **a** Propagation of EMW through shielding material, **b** scattering of EMW in porous material

antennas for transmitting and receiving electromagnetic signals. The VNA records amplitude and phase data during transmission and reception, enabling the calculation of real ($\varepsilon'$) and imaginary ($\varepsilon''$) components of permittivity. The real component of permeability was found to be 1, and the imaginary component was determined to be 0, indicating that the material is non-magnetic [10]. The real part of permittivity ($\varepsilon'$) shows its variable magnitude from 3.25–3.48 with frequency range over 8.2–12.4 GHz. In the frequency range 8.2–9 GHz the value is decreasing rapidly and from 9 to 10.5 GHz there is nonlinear variation with multiple peaks which indicates the various interaction of EM wave with material composition. The interaction of EM wave leads to existence of variety of polarization like dipolar, atomic and ionic as shown in Fig. 6. The rapid decrease in dielectric value with increase of frequency from 8.2 to 9 GHz is due to reduction of space charge polarization effect such as dipolar polarization [24]. With the incident of alternating EM field, the nonpolar molecules are undergoing displacement quickly due to breakage of bonding between the molecules for which the dielectric constant decreases suddenly as there is no intrinsic dipole. Once this polarization takes place due to creation of large numbers of dipoles the electrons gathering takes place which leads to a relaxation process under the action of EM field. This absorbs the EM energy as shown by constant dielectric value for frequency 8.5–9 GHz contributed by ionic polarization. Further, the dielectric constant varies nonlinearly with atomic polarization beyond the frequency 9 GHz followed by atomic polarization, dipolar polarization and ionic polarization similar to that of low frequency region [25]. This happens due to fact that, under the influence of external charge the electron cloud shifts rather than centered on the nucleus which increases the polarizability due to increase of the volume of the electron cloud. This frequency dependent non-linear behavior of dielectric constant causes the power loss at higher frequency, which finds its application in devices where low dielectric constant value required.

The energy loss in form of heat inside the material is basically caused by different type of polarization quantified by dielectric loss ($\varepsilon''$) under the interaction of alternating EM field. For an ideal dielectric material, the real ($\varepsilon'$) and imaginary ($\varepsilon''$) part are always equal to 1 and 0, respectively indicates that the material neither store the energy nor dissipate. But the values of $\varepsilon''$ is less than one, for the synthesized material which shows that it can store the energy but able to dissipate stored energy [26]. As the composition of the dielectric material determines the dielectric value, for larger loss of energy in the material should have larger imaginary part. Thus, based on experimental values it is significant to confirm that the material with thickness 3 mm have the potential to absorb the EM wave. The frequency-dependent loss of the material's dielectric properties decreases with increase of frequency as shown in Fig. 7 and the behavior is repeated within the X-band frequency range. This peculiar behavior of the material governed by two factor such as conduction loss and polarization loss of organic fiber-based absorber [27]. The conductivity loss may be due to the cross linking of both homogeneous and heterogeneous fiber components in hybrid composite which forms easy mobility of electrons. Similarly, the polarization loss is contributed by different polarization such as dipolar polarization and interfacial polarization at low frequency followed by relaxation phenomena and at high frequency ionic and electronic polarization [28, 29]. Further, the factors like incident EM field, temperature, moisture content and density of composition influence the mobility of charge carrier and hence the loss mechanism inside the material. The larger the dielectric constant value lowering the resonant frequency making narrow bandwidth. Further, variation of tangent loss (tan $\delta$)

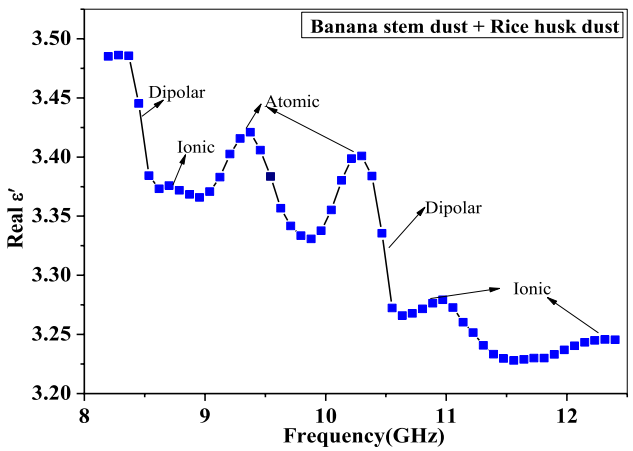



**Fig. 6** Real part of permittivity of BNN/RH hybrid composite with frequency

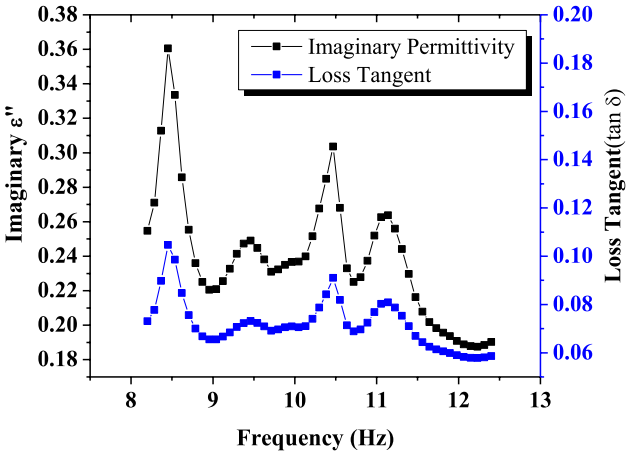



**Fig. 7** Variation of imaginary part of permittivity and tangent loss of BNN/RH hybrid composite with frequency

with frequency as shown in Fig. 7 measured with dielectric loss values which is consistent with dielectric loss because both phenomena involve internal friction, molecular interactions, and dissipative processes contributing to energy loss. The variation indicates that, more is the dielectric loss larger is the tan δ and it decreases with frequency till the relaxation phenomena is reached. The frequency at which the relaxation process reached it drops sharply leading to orientational mechanism which well explains the behaviors of dielectric loss as the frequency increased [18].

Attenuation is a surface phenomenon which attenuates the intensity of EM wave along its length of propagation depending upon the nature of the material. The attenuation coefficient ($\alpha$) shows an increasing trend from 1.71 to 1.81 with increase of frequency from 8.44 to 10.44 GHz and this further shows a linear increasing trend from 1.16 to 1.70 with increase of frequency 8.19–11.14 GHz if the intermediate values are ignored within the frequency 10.5 GHz which is suitably consistent to matching of impedance of the BAN/RH hybrid composite. This variation of α imitates the trend of tan δ confirming the larger the value of tan δ, greater is the attenuation as shown in Fig. 8. It is clear that the value of α has nonlinear trend may be contributed due to heterogeneous composition of material with epoxy polymer [11]. This non uniform distribution may create many skeletal structures with large no. of defects sites. These defects presents on the BAN/RH hybrid composite traps the charge carrier and also break the functional group by unbalancing the charge distribution that leads to reduction of intensity of EM energy on the surface of the hybrid composite followed by the equation, $I = I_0 e^{-\frac{\alpha}{\sqrt{f\mu\sigma\pi}}}$ with $I_0$ is the initial EM wave intensity [12]. Since the synthesized material is lossy dielectric such behaviors contributed by the frequency dependent dispersion between the heterogeneous composition of the BAN/RH hybrid composite. Since at low frequency the BAN/RH hybrid composite inhibits high dielectric constant, the EM wave propagate with low speed indicating high refractive index due to space created by the electron clusters. This leads to decrease in wave length of the refracted wave inside the BAN/RH hybrid composite more rapidly and significantly enhancing the attenuation phenomena as shown in Fig. 8

The effective conductivity of the material shows a nonlinear variation as shown in Fig. 9, consistent with the loss tangent confirmed from the Eq. (8) which is directly dependent on tan$\delta$. The dipoles at low frequency responds the varying EM field and have the loss usually low. When the respond of dipoles falls behind applied EM field with the increase in frequency, loss increases with decrease of stored energy same as phase lag between voltage and current wave. The conductivity ($\sigma$) varies from 1.00 to 1.49 S/m at 8.94 GHz. Directing further increase of $\sigma$ from 1.011 to 1.461 S/m with increase of frequency from 8.94 to 9.47 GHz. Moreover, the value of σ attains the highest value of 2.268 S/m at 10.27 GHz with a sharp drop to 1.23 S/m at 11.49 GHz moving towards a constant value of conductivity of 1.07 S/m from frequency range of 11.49–12.38 GHz. This fluctuated values of $\sigma$ contributed mainly by the conductive loss of EM wave propagate through the material. At low frequency when the EM wave strikes the material it has high dielectric constant which gradually undergoes different polarization leading to transmission of EM wave converted into electric current. The induced electric field facilitates the mobility of charge carrier influenced by the instinct resistance and produces heat followed by consumption of EM energy. Though there is no particular rule for variation of conductivity in microwave absorber under the influence of alternating EM filed still it can be well understood by two mechanisms such as electron migration and electron hopping. Electron

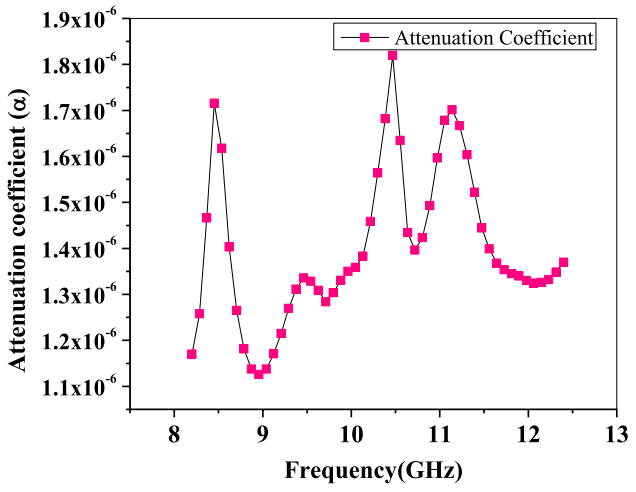



**Fig. 8** Attenuation coefficient of BNN/RH hybrid composite with frequency

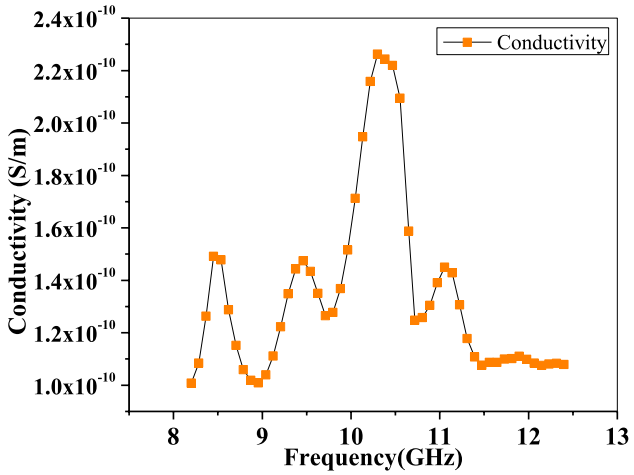



**Fig. 9** Conductivity of BNN/RH hybrid composite with frequency

migration corresponds to electron free movement during propagation of EM wave [13]. Electron hopping refers to transfer of electrons between the interfaces, defects and from one component to other components of the composition of the BAN/RH hybrid composite. As the material is lossy dielectric here the filling ratio in composite affects the conductive network and accordingly the electron hopping controlled by energy barrier restriction. Since different dielectric polarization comes into existence depending on the dielectric constant of the material with non-uniform distribution of the fibrous constituents blended by epoxy polymer the electron hopping may not continuous in all defects for which the effective conductivity shows a fluctuation in its magnitude with the frequency of EM wave [21].

Figure 10 represents the trend of reflection loss (RL) in BAN/RH hybrid composite in the frequency range 8–12 GHz. The fabricated material shows a decreasing trend of RL from 8 to 10 GHz and an increasing trend of RL at higher frequency. The curves shown in the graph were computed based on the real values of the relative dielectric permittivity ($\varepsilon^*$) and permeability ($\mu^*$). MAM are to be considered excellent if they possess RL value below − 10 dB over a wide range of frequencies for an optimal thickness. The evolution of the morphology of flexible EM wave absorbers, as shown in Fig. 10, results in a transition from a shielding function to an absorption function [30]. This is due to an improved impedance match characteristic, which allows for better absorption of EM waves. The mechanism of microwave absorption can be discussed by considering that, when microwave radiation penetrates into a material, it interacts with the sub atomic particles of the fibers, causing energy to be lost as heat through a series of successive interactions. The micro size of the particles in BAN/RH fibers enhances the coupling action at the interface between two fibers. The improved absorption of EMW is due to reduction in particle size to the micron level and can be significantly contributed by quantum size effect. When particles are of micro crystallites, there is wide separation in energy levels. Electrons absorb energy when they move from one energy level to another. This absorption of energy leads to an increase in attenuation, as well as effects on the reflection loss. Due to interfacial polarization associated with size of the dust particle of organic materials this results broadening of band width of reflection loss curve in microwave frequency range. In a general, this sample shows broadband absorber behavior at − 10.4 dB reflectivity with a wide band gap of 2.13 GHz. The absorption bandwidth refers to the frequency range where the intensity of EMW is attenuated by 70% or where the power of the wave is attenuated by 90% [31, 32]. Another main factor is the presence of binding agent in the composite. The epoxy resin used in the BAN/RH/epoxy composite is transparent to EMW and acts only as binding agent [5]. The composite's ability to attenuate power is inherited from the NaOH-treated BAN/RH. This results in a reflection loss of over 90%, making the composite material an excellent choice for designing stealth materials.

Figure 11 displays the impedance matching associated with the BAN/RH hybrid composite material. In other words, it shows how well the material matches the impedance of its surroundings and how much it reduces the strength of a signal passing through it. Generally, for an absorbing material to perform well, its impedance matching should be greater than 0.3. The closer the impedance matching is to 1, the better the absorbing material will perform in terms of matching the impedance of its surroundings [33]. The trend of peaks in Fig. 12 is quite similar to Fig. 8. The optimal performance of electromagnetic absorbers (EMAs) depends on finding a balance between their ability to match

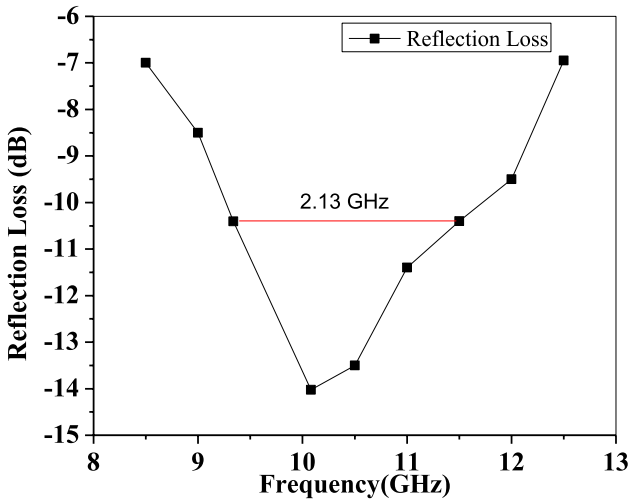



**Fig. 10** Trend of reflection loss (RL) in BAN/RH hybrid composite

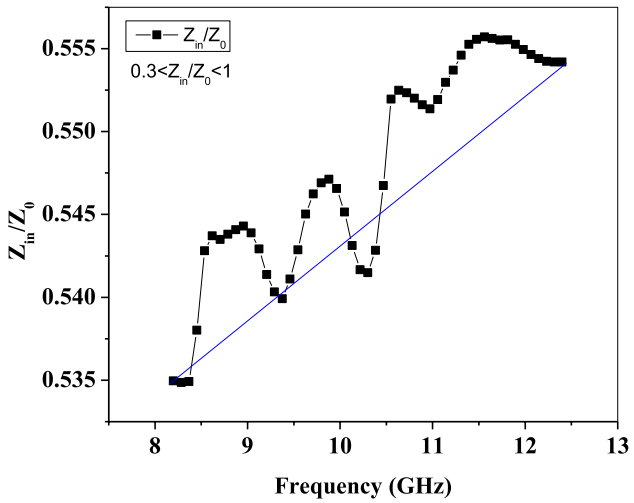



**Fig. 11** Impedance matching of BAN/RH hybrid composite

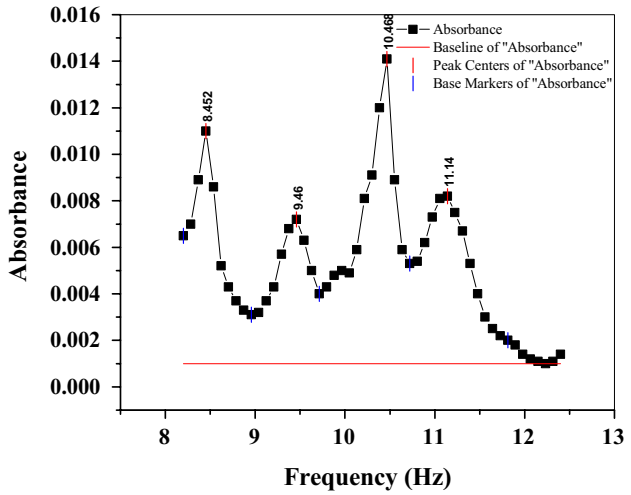



**Fig. 12** Absorbance of BAN/RH hybrid composite

the impedance of the surrounding environment and their ability to strongly attenuate electromagnetic waves. To be considered ideal electromagnetic absorbers, materials must exhibit appropriate impedance-matching properties as well as strong attenuation capabilities. After a thorough comparison, it was found that the BAN/RH hybrid composite material has higher attenuation constant and better impedance matching lies in between $0.3 < Z_{in}/Z_0 < 1$ in X-band. This means that the material was able to effectively absorb electromagnetic waves within this frequency range while also matching the impedance of its surroundings to a satisfactory level. Referring to Fig. 2, when the inter-atomic spaces within a dielectric composite material are replaced with air, which has no electromagnetic (EM) wave attenuation capacity, the overall loss of EM waves within the absorber material is reduced. So conductive materials that have a pore structure not only partially enhance impedance characteristics but also induce interfacial polarization, both of which can increase the ability of the material to absorb EM waves.

Figure 12 describes the frequency dependent absorbance graph of BAN/RH composite within X-Band. From the graph it is observed that at frequency 8.45, 9.46, 10.46, 11.14 Hz, the material exhibits the highest EM absorbance due to the resonance of EM waves within the highly elastic structure of BAN/RH composite which leads to increase in dielectric loss.

Table 1 shows the full width at half maxima (FWHM) values of the absorbance peaks which is calculated using the Eq. (12) in Origin PRO8.5 software.

$$\text{FWHM} = f_2 - f_1 \tag{12}$$

where $f_2$ and $f_1$ are the frequency at which absorbance is half of the maximum peak value. The values represent the bandwidth of the absorbance peaks. The presence of distinct peaks of varying FWHM value suggests the involvement of multiple absorption mechanisms within the material. This multimodal behavior enhances the material's ability to interact with EM waves through various polarization methods as shown in Fig. 6.

**Table 1** FWHM of the absorbance peak of BAN/RH hybrid composite

| Sl. no. | Peak number | Frequency (Hz) | FWHM |
|---|---|---|---|
| 1 | 1 | 8.45 | 1.14 |
| 2 | 2 | 9.46 | 0.45 |
| 3 | 3 | 10.46 | 0.39 |
| 4 | 4 | 11.14 | 0.71 |

## 4 Conclusion

The present work is centered on the synthesis and characterization of a novel hybrid composite material comprising banana stem (BAN) and rice husk (RH) dust, designed for microwave absorption applications. The fabrication process involved a very scientific method for a 50–50 wt.% composition of BAN and RH composite beginning with NaOH treatment, ultrasonication, and subsequent incorporation into an epoxy polymer matrix using a hand layup method. Comprehensive analyses utilizing scanning electron microscopy (SEM), energy-dispersive X-ray spectroscopy (EDS), and Fourier-transform infrared spectroscopy (FTIR) proved the successful elimination of impurities, creating voids and porous skeletal structures with enhanced absorption properties. The computed microwave parameters, including attenuation constant, conductivity, reflection loss, tangent loss, and matched impedance, collectively reflected the dielectric properties of the BNN/RH hybrid composite. The optimal impedance matching value (0.55) fell within the range of $0.3 < Z_{in}/Z_0 < 1$, facilitating efficient EM wave propagation through the absorber with high attenuation (1.63 dB at 11 GHz). The non-uniform distribution of reinforced organic fibers within the epoxy polymer matrix contributed to dielectric and tangent losses, making the composite a promising microwave absorber. The attenuation constant exhibited fluctuations, attributed to the composite's heterogeneous composition. Reflection loss (RL) analysis showcased broadband absorption characteristics, decreasing 90% reflectivity, positioning the composite as a feasible material for microwave absorber applications. Impedance matching confirmed the material's suitability for electromagnetic wave absorption. Absorbance investigations revealed distinct peaks with varying full width at half maxima (FWHM) values, signifying the engagement of multiple absorption mechanisms. The

microwave absorber with thickness 3 mm has satisfactory reflection loss of − 14.01 dB and attenuates the incident EM power of 96.02% through the material. Further, the material has also a bandwidth of 2.13 GHz in the X-band frequency range and represents the novelty of the composite for wide application in designing of stealth material for various electronic devices as well as coating material.

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**Author contributions** PPS: data curation, writing—original draft preparation, software. AKD: writing—reviewing and editing. GN: conceptualization, methodology, supervision, and validation.

**Data availability** The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

## Declarations

**Conflict of interest** The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. Financial interests: the authors declare they have no financial interests.

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